UNIVERSIDAD CARLOS III DE MADRID

Master in Aeronautical Engineering — Air Transport

Aircraft Trajectory Optimisation Lab

Foreword

You hold a chief flight dispatching position working for a young, dynamic airline named *Fly me to the Moon (FlyMoon)*. FlyMoon serves various medium-haul Origin-Destination (O-D) city pairs with its Airbus A320 fleet.

To improve operational profitability, the company aims to eliminate inefficiencies in conventional step-climb/cruise/descent profiles by transitioning toward continuous optimal flight profiles. As the first phase of this operational upgrade, you are tasked with developing a numerical trajectory optimization module in Python using **Pyomo** and nonlinear programming (NLP) solvers such as **IPOPT**.

Aircraft Dynamic Model & BADA 3.11 Performance

For cruise flight at constant pressure altitude h_e (FL380) over a flat Earth model without wind, the 3DoF equations of motion simplify to a 1D system along the airway distance $x_e(t)$:

$$\frac{d}{dt}\begin{bmatrix} x_e(t) \\ V(t) \\ m(t) \end{bmatrix} = \begin{bmatrix} V(t) \\ \dfrac{T(t) - D(V(t), m(t))}{m(t)} \\ -T(t) \cdot \eta(V(t)) \end{bmatrix}$$

where:

- $x_e(t)$ is the distance flown along the route [m].
- $V(t)$ is the True Airspeed (TAS) [m/s].
- $m(t)$ is the instantaneous aircraft mass [kg].
- $T(t)$ is the total engine thrust [N].

BADA Fuel Flow & Thrust Limits

The thrust specific fuel consumption $\eta(V)$ [kg/(N · s)] and operational limits are:

$$\eta(V) = \frac{C_{f1}}{1000 \cdot 60} \cdot \left(1 + \frac{V}{C_{f2}}\right)$$

$$T_{\max} = C_{tc1} \cdot \left(1 - \frac{h_e}{C_{tc2}} + C_{tc3} \cdot h_e^2\right)$$

$$\dot{m}_{\min} = \frac{C_{fc3}}{60} \cdot \left(1 - \frac{h_e}{C_{fc4}}\right) \quad \text{[kg/s]}, \qquad T_{\min}(V) = \frac{\dot{m}_{\min}}{\eta(V)}$$

The required reference parameters (extra envelope flight constraints, aircraft performance data, fuel/thrust coefficients, and atmospheric conditions) must be extracted by the student from the provided BADA 3.11 manual/files and standard atmospheric models (ISA).

Laboratory Assignments & Questions

1: Unidimensional cruise trajectory with fixed arrival time

- Formulate and solve the 1D minimum fuel cruise problem for an Airbus A320 flying 1,000 km at FL380 with initial conditions $V(0) = 420\,\text{kts}$, $m(0) = 51\,000\,\text{kg}$, and fixed arrival time $t_f = 4700\,\text{s}$.
- Implement two separate Pyomo codes using:
  (a) Trapezoidal Direct Collocation.
  (b) Hermite-Simpson Direct Collocation.
- Run grid refinement tests for $N \in \{50, 100, 150, 300\}$ nodes. Plot and discuss the convergence of different solution variables (such as total fuel burn, velocity profiles, and control thrust profiles, ...)
- Include a sensitivity analysis based on different cases (boundary conditions, path constraints, ...)

2: Cost Index sensitivity & Free final time

- Free the final time t_f and replace the objective functional with the Direct Operating Cost (DOC):
$$\min_{x,u,t_f} J_{\text{DOC}} = CI \cdot t_f + (m(0) - m(t_f)) \cdot \left(\frac{1}{100}\right)$$
where $CI \in \left[\frac{\$/\text{hr}}{\$/\text{lb}}\right]$ represents the airline Cost Index.
- Solve the problem for $CI \in \{0, 10, 50, 200, 999\}$.
- Analyze and compare the trade-off between flight time t_f and fuel mass burned. Explain the physical behavior of speed $V(t)$ and the rest of the state variables as CI increases.
- Include a sensitivity analysis based on different cases (boundary conditions, path constraints, ...)

3: 2D Horizontal Optimal Trajectory

- Formulate a 2D horizontal flight path (λ_e, θ_e) at FL380 ($h_e = \text{const}$, $\gamma = 0$) over a spherical Earth model (R_e) between 2 different European airports.
- Governing equations of motion:
$$\frac{d}{dt}\begin{bmatrix}\lambda_e \\ \theta_e \\ V \\ \chi \\ m\end{bmatrix} = \begin{bmatrix}\frac{V(t)\cos\chi(t)+w_x}{R_e\cos\theta_e(t)} \\ \frac{V(t)\sin\chi(t)+w_y}{R_e} \\ \frac{T(t)-D(V,m,\mu)}{m(t)\cdot} \\ \frac{g\cdot\tan\mu(t)}{V(t)} \\ -T(t)\cdot\eta(V)\end{bmatrix}$$
- Aerodynamic Relations ($\gamma = 0$):
$$L = \frac{m \cdot g}{\cos\mu}$$
- Compare the zero-wind ($w_x = w_y = 0$) optimal trajectory against the Orthodromic (Great Circle) path.

- Introduce constant wind components (w_x, w_y) and evaluate lateral path deviations and fuel impact.
- Include a sensitivity analysis based on different cases (boundary conditions, path constraints, ...)

4: 2D Vertical Optimal Trajectory

- Formulate a 2D vertical flight profile (x_e, h_e) over a flat Earth allowing continuous climb, cruise, and descent phases along a fixed track ($\chi = \text{const}$, $\mu = 0$).
- Governing Equations of Motion:

$$\frac{d}{dt}\begin{bmatrix} x_e \\ h_e \\ V \\ \gamma \\ m \end{bmatrix} = \begin{bmatrix} V(t)\cos\gamma(t) \\ V(t)\sin\gamma(t) \\ \frac{T(t)-D(h_e,V,C_L)-m(t)g\sin\gamma(t)}{m(t)} \\ \frac{L(h_e,V,C_L)-m(t)g\cos\gamma(t)}{m(t)\cdot V(t)} \\ -T(t)\cdot\eta(V) \end{bmatrix}$$

- Analyze how optimal cruise altitude and speed profiles shift when varying the Cost Index (CI).
- Include a sensitivity analysis based on different cases (boundary conditions, path constraints, ...)